RESPONSE TO REVIEWER COMMENTS

We thank the reviewers for reading the manuscript carefully and for providing detailed comments. Their questions showed us where the scope and technical contributions needed clearer explanation. Because the rebuttal has a strict length limit, this letter provides the full experimental tables and implementation details that could not fit there.

I. RESPONSE TO REVIEWER 1

R1-O1. *Several components of XOver, such as retrieval-augmented prompting, submodular exemplar selection, parallel sampling, and execution-guided refinement, build upon existing ideas. The primary novelty lies in their adaptation and integration for multilingual Text-to-OverpassQL rather than fundamentally new algorithmic techniques.*

Response. Thank you for raising the question of where the technical contribution lies. The individual mechanisms have precedents; our contribution is their formulation for multilingual Text-to-OverpassQL. It includes MOSMNL, XEC, an OverpassQL-specific execution-grounded utility, cross-lingual set coverage, the NP-hardness and submodularity results, and an approximation-guaranteed greedy algorithm. These parts optimize the same downstream objective $\mathcal{F}_\ell$. Retrieval approximates this black-box objective through grounding and coverage, the selected context determines which programs are sampled, and execution-guided voting and Revision calibrate those candidates using their execution behavior. Inference continues the optimization begun by cross-lingual contextualization.

R1-O2. *The proposed retrieval utility combines semantic similarity with an angle-based grounding score. While the empirical improvements are convincing, the paper could better explain why this formulation is preferable to alternative retrieval objectives and whether the gains mainly come from the angle-based representation or the supervision strategy used to construct the grounding labels.*

Response. The gain comes from both the supervision and the retrieval geometry. Cosine preserves broad semantic relevance, while the angular term separates program-critical relations among otherwise similar questions. For example, “找学校附近的餐厅” (restaurants near schools) and “Find restaurants in schools” have cosine similarity 0.847 under Qwen3-Embedding-0.6B, although one requires proximity and the other containment. In the matched evaluation in Table I, Angle+Cosine reaches 76.13 average OQS, compared with 75.40 for Angle and 74.72 for Cosine. The labels matter independently: they raise the CrossEncoder result from 71.80 to 73.03.

R1-O3. *The proposed framework introduces multiple stages, including retrieval, greedy exemplar selection, parallel generation, execution-based voting, and iterative revision. Although effectiveness is demonstrated, the paper provides limited discussion regarding the additional inference latency and computational overhead introduced by these components.*

Response. Table II reports measurements on 400 development queries (50 per language) using deepseek-v4-flash. CosSim ICL makes one model call, uses 648.3 tokens, and executes once per query; its p50/p95 latency is 2.578/6.969 seconds. Full XOVER makes eight parallel calls, uses 6,014.5 tokens, and executes

TABLE I: Retriever attribution and training-label ablations.

(a) Retriever comparison

Retriever	Grounding label	EN	ZH	YUE	FR	DE	JA	KO	RU	Avg.
CrossEncoder [1]	✗	73.22	71.50	71.10	71.58	72.45	71.05	71.04	72.42	71.80
CrossEncoder [1]	✓	74.46	72.61	72.31	73.61	73.39	72.38	72.43	73.08	73.03
SimCSE [2]	✓	75.18	73.46	73.38	74.24	74.28	72.80	73.26	73.59	73.77
Cosine Objective	✓	76.75	74.51	74.24	74.41	75.32	73.92	73.83	74.74	74.72
Angle Objective	✓	77.07	74.88	75.13	75.69	75.96	74.99	74.74	74.74	75.40
Angle & Cosine Objective	✓	**78.22**	**75.73**	**75.80**	**76.27**	**76.40**	**75.44**	**75.37**	**75.81**	**76.13**

(b) Grounding-label sensitivity

Training labels	Spearman ↑	Pearson ↑	MSE ↓	nDCG@6 ↑
Full	**0.441**	**0.481**	**0.075**	**0.846**
w/o Tag	0.297	0.334	0.087	0.798
w/o Skeleton	0.436	0.473	**0.075**	0.842
w/o OQS utility	0.299	0.318	0.112	0.804

nine times per query. Its p50/p95 latency is 6.285/30.595 seconds, or 2.44×/4.39× that of CosSim. On the critical path, warm selection over precomputed scores takes 4.8/6.2 ms, parallel generation 2.244/3.153 seconds, and candidate execution 3.657/25.797 seconds. These figures come from one run. Warm selection excludes the online encoder and scorer, and absolute execution time varies with endpoint load and cache state.

TABLE II: Inference overhead on 400 development queries (50/language) with deepseek-v4-flash. Latencies are one-run stage-accounting estimates; warm selection excludes online encoder/scorer cost.

System	p50 (s)	p95 (s)	Calls/q	Tokens/q	Exec./q	Rel. p50
CosSim ICL	2.578	6.969	1	648.3	1	1.00×
XOver-Greedy	2.478	5.716	1	752.5	1	0.96×
XOver-Vote	6.039	28.463	8	6,014.5	8	2.34×
Full XOVER	6.285	30.595	8	6,014.5	9	2.44×

R1-D1. *The paper would benefit from a more detailed comparison with recent learned retrieval or reranking methods beyond cosine-similarity retrieval to better position the proposed grounding model.*

Response. To answer this question, we added learned CrossEncoder and SimCSE controls under the same generator, exemplar pool, shot count, and development set. Table I shows 71.80 average OQS for CrossEncoder without grounding labels, 73.03 for the same model with the labels, and 73.77 for supervised SimCSE. Angle+Cosine with the same grounding supervision reaches 76.13. Thus, a learned reranker alone does not account for the result.

R1-D2. *The contribution of each execution-related component (parallel generation, execution-guided voting, and iterative revision) could be further isolated through additional ablation studies.*

Response. Table III separates candidate generation, voting, execution guidance, and Revision. On the held-out test set, Non-EN EX is 39.3 for greedy decoding with $N = 1$, 42.4 for majority voting with $N = 8$, and 44.4 for execution-guided voting over the same candidate pool. Oracle@$N = 8$ is 54.6. Sampling therefore creates 15.3 points of oracle headroom; majority voting recovers 3.1 points over greedy decoding, and execution guidance adds another 2.0. After voting, 71 of the 7,000 protocol-matched Non-EN outputs still hard-fail. Revision reduces this number to 14, an 80.3% reduction. Its main role is to recover residual execution failures, not to correct semantics broadly.

TABLE III: Completed execution-component ablations. Panel (a) uses the held-out test set. Panel (b) is a separate controlled development-set factorial with 1,000 aligned queries/language, deepseek-v4-flash, and the same 120-s client cap in all cells. Values are percentages unless noted.

(a) Candidate generation and selection

Strategy	EN EX	Non-EN EX	EN EX_{soft}	Non-EN EX_{soft}
Oracle@N=8	58.3	54.6	70.6	65.8
Greedy (N=1)	43.7	39.3	60.1	54.7
Majority (N=8)	46.5	42.4	63.4	58.4
Execution-guided (N=8)	**47.7**	**44.4**	**65.3**	**61.4**

Setting	EX ↑		EX_{SOFT} ↑		EX_{ERROR} ↓		Revision Success ↑	
	EN	Non-EN	EN	Non-EN	EN	Non-EN	EN	Non-EN
Greedy w/o Revision	46.1	44.7	59.5	57.4	62	52.6	–	–
Greedy + Revision	46.4	45.0	60.4	58.1	23	19.1	**22**	**15.4**
Execution-guided w/o Revision	**50.6**	48.9	**66.1**	63.2	4	10.1	–	–
Execution-guided + Revision (Ours)	**50.6**	**49.0**	**66.1**	**63.3**	**1**	**2.3**	0	2.1

R1-D3. *The construction of the grounding labels combines multiple heuristic signals, including tag overlap, skeleton similarity, and execution quality. More discussion on the sensitivity of these design choices would improve the reproducibility and interpretability of the retrieval model.*

Response. Table I reports Spearman 0.441 and nDCG@6 0.846 with full supervision. Removing Tag lowers them to 0.297 and 0.798; removing OQS utility lowers them to 0.299 and 0.804. Removing Skeleton has little effect (0.436 and 0.842). Under this one-seed protocol, most of the supervision signal comes from Tag overlap and OQS-derived generation utility. We will also clarify that OQS utility measures generation quality, not execution fidelity directly.

R1-D4. *Since MOSMNL is likely to become an important benchmark, additional analysis of annotation quality, translation consistency, and potential benchmark limitations would further strengthen the dataset contribution.*

Response. MOSMNL uses a three-stage human-in-the-loop process. Google Translate produces the initial version. Place mentions are also translated separately and compared with the full sentence; the authors inspect any discrepancy, using GPT-5.1 only for post-editing. Next, GPT-5.1, GPT-4o mini, and Gemini-3-Flash-Preview rate each item independently on six dimensions: semantic fidelity, fluency, terminology,

cultural appropriateness, task-semantic preservation, and structural integrity. An item is flagged when at least two models give a low rating on a core dimension. Every flagged item is then reviewed and, when needed, revised by a person. Human decisions determine the released text. Authors with native-level proficiency reviewed Mandarin, Cantonese, and Korean. Undergraduate or graduate language specialists reviewed French, German, Japanese, and Russian; they were recruited through Xianyu and paid RMB 2 per item.

The benchmark is parallel and translation-derived from English OverpassNL. This keeps the gold program and query complexity fixed across languages. It also means that the benchmark does not represent organically authored, language-specific requests and inherits the source corpus's coverage of tags, operators, query complexity, and geography. We will state these limits directly.

II. Response to Reviewer 3

R3-O1. *The paper does not convincingly establish that multilingual Text-to-OverpassQL constitutes a sufficiently distinctive and consistently severe challenge beyond existing cross-lingual semantic parsing and structured-query generation problems.*

Response. Thank you for asking us to define the scope more carefully. We make a narrow claim: the difficulty studied here is grounding a non-English question in one English exemplar pool. We do not claim that multilingual Text-to-OverpassQL is harder than every cross-lingual semantic-parsing task. A useful exemplar must provide the OSM tags, spatial operators, and query structure needed by the generator. OSM tags are English-centric, place names vary across languages, and semantically close relations such as "near" and "in" may require different programs. Broad semantic similarity can therefore retrieve the right intent but the wrong programmatic evidence. XOver tests whether exemplar contextualization alone can make one English pool work across eight languages without changing the generator.

R3-O2. *The contribution and representativeness of MOSMNL are limited by its translation-derived construction, weakly justified language selection, and close dependence on the English OverpassNL corpus.*

Response. This comment helped us state the benchmark's contribution and limits more precisely. MOsmNL is a parallel extension of OverpassNL: aligned questions share the same gold OverpassQL program and geographic scope, so program difficulty remains fixed in cross-language comparisons. It supports controlled multilingual evaluation for a previously English-only task, but it does not cover organically authored, language-specific requests. The languages reflect the qualified reviewers available to us; we do not claim exhaustive typological coverage. Authors with native-level proficiency reviewed Mandarin, Cantonese, and Korean. Undergraduate or graduate specialists reviewed French, German, Japanese, and Russian; they were recruited through Xianyu and paid RMB 2 per item. We will state this selection criterion and the dependence on OverpassNL in the revision.

R3-O3. *The technical contribution of XEC is unclear, as the framework combines familiar exemplar retrieval, diversity-aware subset selection, parallel sampling, and execution-guided filtering through a complicated surrogate formulation whose task-specific benefit is limited.*

Response. Thank you for giving us the opportunity to clarify the role of XEC. Its black-box target is downstream generation quality $\mathcal{F}_\ell$, which cannot be evaluated for every candidate subset at practical cost. XEC replaces that search with a tractable surrogate based on semantic proximity, OverpassQL-specific grounding, and set coverage. The selected context defines the programs explored by parallel sampling. Voting and Revision then use execution behavior to calibrate those candidates against the same downstream objective. The NP-hardness result and approximation guarantee concern the retrieval surrogate; inference completes the optimization after contextualization.

R3-O4. *The experimental analysis does not clearly separate improvements from multilingual grounding, general exemplar-quality modeling, multi-candidate generation, and language-independent execution post-processing.*

Response. We separate the contributions with controlled comparisons, while treating retrieval and inference as parts of the same objective. With every other component fixed, Full8x improves OQS over the English-only grounding model for all seven non-English languages, by 3.52 points on average; the English gain is 2.16 points. The difference between these gains is 1.36 points (95% CI $[0.64, 2.08]$), which isolates the additional benefit of multilingual grounding. Table III then separates sampling, voting, and Revision. These inference steps improve both EN and Non-EN because they use language-invariant execution behavior to calibrate $\mathcal{F}_\ell$.

R3-D1. *The motivation combines cross-lingual lexical mismatch, exemplar redundancy, and discarded execution feedback, but does not sufficiently establish why this combination is uniquely difficult for OverpassQL. Multilingual semantic parsing benchmarks already study language-independent target programs, while exemplar redundancy and execution-guided generation are general problems in in-context program synthesis. The OverpassQL-specific motivation is primarily illustrated through a small number of OSM-tag and spatial-preposition examples.*

Response. Thank you for asking us to separate the general program-synthesis mechanisms from the OverpassQL-specific difficulty. Exemplar redundancy and execution guidance are general issues. The OverpassQL-specific problem is that an English exemplar must supply the tags, spatial operators, and query structure required by a non-English question. Cross-lingual supervision handles this grounding boundary. Coverage avoids repeating the same functional evidence, and execution feedback calibrates the programs generated from that context. The Full8x comparison in R3-O4 provides controlled evidence that multilingual grounding adds a Non-EN benefit beyond the motivating examples.

R3-D2. *The claimed cross-lingual adaptation gap is strongly dependent on the selected model. In Table II, translation adaptation improves the non-English mean OQS of DeepSeek-V4-Flash by only 1.0 point and GPT-4o-mini by 2.0 points, with negligible differences on several languages. The large gap is mainly observed for the smaller OsmT system, weakening the claim that translation adaptation is broadly insufficient for modern multilingual LLMs.*

Response. Thank you for pointing out the model dependence. The gap is larger for English-trained OsmT and smaller for multilingual LLMs and CosSim with a multilingual embedding model, so Table II should

not be read as claiming a uniformly large gain from translation adaptation. On the aligned development set, the same 6-shot DeepSeek-V4-Flash configuration reaches 71.9 EN OQS and 70.7 average Non-EN OQS when each language has a translated exemplar pool. XOVER uses one English pool for all languages and reaches 77.3 and 76.2, respectively. This is a 5.5-point Non-EN improvement over the strongest translation-adapted baseline without constructing per-language pools. Full8x also improves on English-only grounding by 3.52 Non-EN points and 2.16 EN points; the difference is 1.36 points (95% CI $[0.64, 2.08]$).

R3-D3. *Figure 2 occupies substantial space but has low technical information density. It visualizes the four-stage pipeline through generic icons and labels, while omitting the grounding utility, coverage kernel, set-selection objective, and execution-voting criterion that constitute the main technical details of XOver.*

Response. We will revise Figure 2 to show the grounding utility, coverage kernel, set-selection objective, candidate generation, execution outcomes, and voting criterion. Decorative elements will be reduced. The revised figure will connect the mathematical formulation to the four executable stages instead of simply listing the stages.

R3-D4. *MOSMNL is largely constructed from machine-translated and human-verified variants of English source questions, with the language variants sharing the same gold OverpassQL programs. The choice of Mandarin, Cantonese, French, German, Japanese, Korean, and Russian is described as typologically diverse but is not supported by a clear selection criterion. The paper also provides limited information about native-language query phenomena, annotator backgrounds, agreement, and language-specific translation quality.*

Response. Sharing gold programs keeps query semantics and complexity fixed across languages, but it does not reproduce the distribution of organically authored native-language queries. We will state both consequences. As noted in R3-O2, qualified reviewer availability determined the language set; it was not chosen to provide exhaustive typological coverage. The revised construction section will describe the three-stage review process, six rating dimensions, two-model flagging rule, manual adjudication, and reviewer qualifications. It will also state that the French, German, Japanese, and Russian specialists were recruited through Xianyu and paid RMB 2 per item. The present benchmark does not establish coverage of language-specific phenomena.

R3-D5. *The benchmark contribution is characterized mainly through task type, construction source, dataset size, and number of languages. Table I does not establish distinctive coverage in terms of OSM tags, spatial operators, query complexity, linguistic phenomena, or geographic reasoning. The evaluation also does not include a direct test-time translation pipeline that translates the target question into English and then applies an English Text-to-OverpassQL system.*

Response. The benchmark preserves gold programs to support controlled multilingual alignment. It does not expand the structural coverage of the English source corpus. We will state that it inherits OverpassNL's distributions of OSM tags, spatial operators, query complexity, and geographic reasoning. Translation-derived questions also cannot establish coverage of native-language phenomena.

For the requested direct baseline, Table IV applies Google Cloud Translation before the unchanged six-shot backbone on 1,000 test questions per language. Test-time Translate obtains 68.93 OQS, 34.73 EX, and 49.60 EX_{soft}; XOVER obtains 71.85, 44.40, and 61.52. XOVER uses no test-time translation and has a larger advantage on the execution-based metrics.

TABLE IV: Comparison of retrieval and translation baselines. Scores are percentages averaged over the seven non-English languages.

Method	Dev OQS	Test OQS	Test EX	Test EX_{soft}
Same-language CosSim	70.72	70.41	39.26	54.60
Test-time translation (EN)	69.50	68.93	34.73	49.60
XOVER	**76.16**	**71.85**	**44.40**	**61.52**

R3-D6. *The theoretical analysis concerns the handcrafted surrogate objective, rather than the actual downstream generation quality. NP-hardness follows from incorporating a log-determinant subset-selection term, and the approximation guarantee applies only to this surrogate after top-m filtering. The coverage component itself has limited empirical impact: removing it decreases non-English OQS by only 0.2 points, despite the extensive theoretical treatment devoted to the XEC formulation.*

Response. Thank you for asking us to separate the surrogate guarantee from downstream generation quality. The theoretical result applies to the computable XEC surrogate after candidate pruning. It gives no guarantee for black-box generation quality. Coverage remains necessary for the set-level objective because Similarity and Grounding score exemplars one at a time. Without Coverage, selection reduces to modular top-k ranking and cannot detect redundant evidence. The DPP log-determinant term instead compares grounding-score profiles over the multilingual query bank. It changes 21% of top-k selections and increases OSM tag-key recall. Removing it lowers Non-EN OQS from 76.4 to 76.2, EX from 45.4 to 44.8, and EX_{soft} from 57.7 to 56.9. Coverage alone is insufficient: it obtains 59.6 OQS, 24.7 EX, and 36.7 EX_{soft} on Non-EN. Its role is redundancy control within a relevant candidate set.

R3-D7. *The performance gains are not clearly attributable to a specifically multilingual mechanism. Execution-guided voting raises execution accuracy from 43.7 to 47.7 on English and from 39.3 to 44.4 on non-English queries, showing substantial gains on both sides. The grounding ablation also affects English and non-English performance similarly. A considerable portion of XOver's advantage therefore appears to arise from general retrieval supervision, multi-candidate sampling, and execution-based selection rather than cross-lingual contextualization alone.*

Response. The Full8x control answers this attribution question while keeping the rest of the system fixed. Relative to English-only grounding supervision, Full8x improves every non-English language, by 3.52 OQS points on average; the English gain is 2.16 points. The difference is 1.36 points (95% CI $[0.64, 2.08]$). Retrieval and inference still act on the same $\mathcal{F}_\ell$ objective: the cross-lingual context determines the candidate programs, and execution-guided voting and Revision calibrate them using language-invariant execution behavior. Improvements on both EN and Non-EN are consistent with this division of labor.

R3-D8. *The execution-failure analysis provides limited evidence about semantic correctness. It aggregates parse failures, empty results, preprocessing failures, connection failures, timeouts, and server-side resource errors, although several categories do not directly reflect the semantic quality of generated programs. The paper defines these error types but does not report their observed distribution. Moreover, revision reduces non-English execution failures from 10.1 to 2.0 per 1,000 queries while increasing execution accuracy by only 0.1 point, indicating that the large reliability gain primarily concerns executability rather than correctness.*

Response. Thank you for drawing a clear line between executability and semantic correctness. Table V reports them separately. Execution-guided voting raises Non-EN EX from 39.3 to 44.4, but 71 outputs still hard-fail. Revision reduces that number to 14, an 80.3% drop: parse errors fall from 48 to 9, Nominatim/preprocessing failures from 12 to 2, and runtime timeouts from 7 to 0. Prediction-empty mismatches are executable but denotationally wrong. Only 5 of the 71 initially hard-failing predictions become EX-correct. Revision mainly repairs residual execution failures; it is not a broad semantic corrector.

TABLE V: Observed Non-EN execution outcomes before and after Revision on 7,000 protocol-matched development queries. Parentheses give counts per 1,000. Empty-result mismatches are executable but denotationally incorrect, and are excluded from hard failures.

Observed outcome	**Before Revision**	**After Revision**
Parse error	48 (6.9)	9 (1.3)
Nominatim/preprocessing	12 (1.7)	2 (0.3)
Runtime timeout	7 (1.0)	0 (0.0)
Server timeout	2 (0.3)	1 (0.1)
Static-regex error	2 (0.3)	2 (0.3)
Hard execution failures	**71 (10.1)**	**14 (2.0)**
Prediction-empty mismatch	282 (40.3)	309 (44.1)

III. Response to Reviewer 4

R4-O1. *The mathematical formulation of the XEC problem may be unnecessarily complex given the marginal empirical benefits of its coverage component.*

Response. Thank you for asking why Coverage appears in the formulation. XEC needs a set-level term because Similarity and Grounding score exemplars independently. Setting $\lambda = 0$ gives modular top-k ranking, which cannot detect repeated evidence within the selected set. Coverage compares each English exemplar's grounding-score profile over the multilingual query bank through a DPP log-determinant objective. It changes 21% of top-k selections and raises OSM tag-key recall. Removing it lowers Non-EN OQS/EX/$\mathrm{EX}_{\mathrm{soft}}$ by 0.2/0.6/0.8 points. We claim a structural role in redundancy control, not numerical dominance.

R4-O2. *The evaluation does not clearly disentangle whether XOver specifically solves cross-lingual challenges or broadly improves general Text-to-OverpassQL generation performance.*

Response. Thank you for asking us to separate the cross-lingual gain from the general inference gain. Retrieval and inference act on the same $\mathcal{F}_\ell$ objective. Retrieval maps questions in eight languages to one English exemplar pool and approximates downstream quality through grounding and coverage. The selected context determines which candidate programs are sampled; voting and Revision then calibrate them using execution behavior. With all other components fixed, Full8x improves over English-only grounding by 3.52 Non-EN OQS points and 2.16 EN points. The additional Non-EN gain is 1.36 points (95% CI $[0.64, 2.08]$).

R4-O3. *The manuscript lacks sufficient methodological details regarding the translation and manual annotation processes required to fully reproduce the MOSMNL benchmark.*

Response. We will expand the construction section to specify the initial translation system, separate place-name checks, six quality dimensions, independent model ratings, the two-model flagging rule, mandatory human review of flagged items, final human adjudication, and reviewer qualifications.

R4-D1. *The XEC formulation is potentially unnecessarily complex. From Table VI, it appears that the Coverage Score Function $\mathcal{C}$ only has a minor effect on Non-EN OQS and no effect at all on EN OQS. If the $\mathcal{C}$ term were removed (i.e., setting $\lambda = 0$), the NP-hardness proof via MESP's membership may not apply, potentially simplifying XEC and the subsequent XOver retrieval algorithm. The paper should explore alternative problem formulations and algorithms for a more elegant solution. Otherwise, the authors should prove (theoretically and/or empirically) that this complex formulation is strictly necessary for a specific set of problem requirements.*

Response. Thank you for drawing attention to the $\lambda = 0$ case. Setting $\lambda = 0$ reduces the problem to modular relevance ranking, and the MESP-based hardness argument no longer applies. The simpler objective also loses the set-level requirement: independent scores cannot detect redundant functional evidence. Coverage models complementarity with grounding-score profiles over the multilingual query bank and a DPP log-determinant objective. It changes 21% of top-k selections and raises OSM tag-key recall. Without it, Non-EN OQS falls from 76.4 to 76.2, EX from 45.4 to 44.8, and $\mathrm{EX}_{\mathrm{soft}}$ from 57.7 to 56.9. The extra formulation is needed for set-level redundancy control; Coverage does not supply relevance by itself.

R4-D2. *It is unclear whether XOver primarily (A) solves the multilingual/cross-lingual problem or (B) generally improves Text-to-OverpassQL. Across the evaluation, XOver improves both EN and Non-EN scores, maintaining roughly the same gap between Non-EN and EN scores as the gap achieved by in-context learning (ICL) with translation (as seen in Figure 1 and Table IV). If (B) is the primary contribution, the authors should consider framing XOver as such, potentially in a separate manuscript, keeping the motivation and MOSMNL benchmark with baseline results in the current manuscript. If (A) is the case, please better highlight how XOver specifically solves multilingual challenges rather than overall Text-to-OverpassQL generation challenges.*

Response. Thank you for asking us to state the primary contribution more clearly. It is (A): serving non-English questions from one English exemplar pool without translating that pool, translating test questions,

or adapting the generator. Cross-lingual contextualization provides the English tags, operators, and structures needed by the question, and the resulting context determines which programs inference explores. Voting and Revision use language-invariant execution behavior to complete the same optimization, so improvements on EN do not make them separate additions. The Full8x control measures the multilingual component directly: +3.52 Non-EN versus +2.16 EN OQS over English-only grounding, a difference of 1.36 points (95% CI $[0.64, 2.08]$).

R4-D3. *Please elaborate on the following details sufficiently to ensure the reproducibility of MOSMNL: 1. How did the pipeline translate 8180 EN NLQs into the 7 Non-EN languages? 2. What specific steps were taken to ensure the correctness of these translations, and how was it determined whether each NLQ was translatable? 3. What was the exact procedure for the "manual annotation"? 4. What were the qualifications of the manual annotators, and what steps were implemented to ensure consistency over tens of thousands of queries?*

Response. We address the four requested items in order. (1) Google Translate generated the initial seven target-language versions. Place mentions were translated separately and compared with the full-sentence output using rule-based checks. (2) The authors inspected the detected discrepancies and used GPT-5.1 only as a post-editing aid. Three models then rated each output independently on semantic fidelity, fluency, terminology, cultural appropriateness, task-semantic preservation, and structural integrity. If at least two models gave a low rating on a core dimension, a person reviewed and, when needed, revised the item. (3) Human revisions determined the released text; "manual verification and correction" describes this stage more accurately than "manual annotation." (4) Authors with native-level proficiency reviewed Mandarin, Cantonese, and Korean. Undergraduate or graduate language specialists reviewed French, German, Japanese, and Russian; they were recruited through Xianyu and paid RMB 2 per item. Shared gold programs, separate place-name checks, a common rating rubric, a deterministic flagging rule, and final human decisions kept the batches consistent.

R4-D4. *What are the results when using Only Coverage $\mathcal{C}$ in Table VI?*

Response. Table VI reports the strict Only-Coverage intervention. We set $\mathcal{A} = 0$ and $\mathcal{G} = 0$, then select one query-independent set of six exemplars from 5,305 deduplicated English candidates. On 1,000 development questions per language, it obtains 60.37/59.58 EN/Non-EN OQS, 27.60/24.74 EX, and 39.27/36.72 $\mathrm{EX}_{\mathrm{soft}}$. This deterministic Bailian `deepseek-v4-flash` run used `retry_errors=false`. The historical Full and without-Coverage rows used an earlier provider route, which confounds fine-grained cross-row comparisons. Coverage cannot replace query-dependent relevance and grounding. It controls redundancy after a relevant candidate set has been formed.

R4-D5. *Typo: "MOSMNLtest set" → "MOSMNL test set" (similar to "MOSMNL dev set").*

Response. We corrected the phrase to "MOSMNL test set" and will run the same spacing check throughout the revised manuscript.

TABLE VI: Strict Only-Coverage results on the development set (1,000 queries/language). The intervention sets $A = 0$ and $G = 0$ and uses one query-independent set of six exemplars selected from 5,305 deduplicated English candidates.

Group	**OQS (%)**	**EX (%)**	**EX$_{\text{soft}}$ (%)**
EN	60.37	27.60	39.27
Non-EN	59.58	24.74	36.72
Eight-language average	59.68	25.10	37.04

SUMMARY OF REVISIONS

The revision will sharpen the one-English-pool motivation and use the Full8x control to measure the additional multilingual benefit. It will explain how retrieval and inference act on $\mathcal{F}_\ell$, and why Coverage is part of the set-level objective. We will also add the requested retrieval, execution, efficiency, translation, Only-Coverage, and failure-distribution results; expand the benchmark protocol and limitations; revise the framework figure; and correct the reported typo.

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